

Gaussian laws are exactly the universal carriers of normalized log-mgf convexity

Candidate full classification for the question in arXiv:2311.08351

17 August 2026

Result. Let X be an $\mathbb{R}^n$ -valued random vector whose linear functionals have finite exponential moments of every real order. The normalized log-moment generating function

$$\lambda \mapsto \frac{1}{\lambda} \log \mathbb{E} e^{\lambda F(X)}, \quad \text{with value } \mathbb{E} F(X) \text{ at } \lambda = 0,$$

is convex on $\mathbb{R}$ for every convex F for which all positive exponential moments are finite if and only if X is a possibly shifted, singular, or deterministic multivariate Gaussian. In fact, necessity already follows by testing affine functions.

1 Source question and precise interpretation

Chen [1] proves that if g is a finite-dimensional Gaussian vector and F is convex, then

$$\Lambda_{F(g)}(\lambda) := \frac{1}{\lambda} \log \mathbb{E} e^{\lambda F(g)}$$

is convex in $\lambda \in \mathbb{R}$, with the continuous value $\mathbb{E} F(g)$ at zero. After the theorem the source asks whether the conclusions hold for *more general random vectors*. The relevant discussion on source PDF page 7 is reproduced below.

1.4 Discussion

Following Theorem 1 it is natural to ask if they hold for more general random vectors. One simple extension is to consider $X = \rho(g)$ for some $\phi : \mathbb{R}^l \rightarrow \mathbb{R}^n$ and g an l -dimensional standard normal. If $F : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $F_0(x) := F(\rho(x))$ is convex on $\mathbb{R}^l$, then the conclusions of Theorems 1, 2, and 3 as well as (4), (5), and (6) hold for $F(X)$. As an example observed earlier in [23, 24, 32], for any $k \geq 1$, if $l = kn$ and $\rho(x) = (\sum_{j=k(i-1)+1}^{ki} x_j^2)_{1 \leq i \leq n}$, then $\rho(g)$ is a vector of n i.i.d. $\chi^2(k)$ random variables. If F is assume to be convex and 1-conditional, i.e., $F(y_1, \dots, y_n) = F(|y_1|, \dots, |y_n|)$, then F_0 is convex.

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Figure 1: The extension question in arXiv:2311.08351, source PDF page 7.

There are two different extensions one might seek. A fixed non-Gaussian scalar random variable can enjoy convexity of its own normalized log-mgf. The direct vector-valued analogue of Chen's theorem, however, asks for a *carrier law*: a law of X for which the conclusion holds after every convex observation F . The theorem below completely classifies those laws.

Assume throughout that

$$\mathbb{E} e^{\langle u, X \rangle} < \infty \quad (u \in \mathbb{R}^n). \quad (1)$$

This is the natural condition making every affine test admissible. For a finite convex $F : \mathbb{R}^n \rightarrow \mathbb{R}$, define

$$\Lambda_{F,X}(\lambda) = \begin{cases} \lambda^{-1} \log \mathbb{E} e^{\lambda F(X)}, & \lambda \neq 0, \\ \mathbb{E} F(X), & \lambda = 0. \end{cases} \quad (2)$$

We quantify only over F such that the positive exponential moments in (2) are finite. Negative moments are then automatic: every finite convex function has an affine lower support, and (1) applies.

2 Proof intuition

Linear functions are convex with both signs. For a direction v , reflection turns its normalized log-mgf into the negative reflection of the original function. Convexity for v and $-v$ therefore makes one function both convex and concave, hence affine. Multiplying by λ shows that every projected cumulant generating function is quadratic, which is precisely Gaussianity. The converse is Chen's theorem after representing a possibly singular Gaussian vector as an affine image of a standard one.

3 The classification

For $v \in \mathbb{R}^n$, let

$$K_v(\lambda) = \log \mathbb{E} e^{\lambda \langle v, X \rangle}, \quad \psi_v(\lambda) = \begin{cases} K_v(\lambda)/\lambda, & \lambda \neq 0, \\ \mathbb{E} \langle v, X \rangle, & \lambda = 0. \end{cases}$$

Lemma 1 (The sign-pair rigidity mechanism). *If both ψ_v and ψ_{-v} are convex on $\mathbb{R}$, then ψ_v is affine and $\langle v, X \rangle$ is Gaussian, allowing zero variance.*

Proof. The cumulant generating functions satisfy

$$K_{-v}(\lambda) = K_v(-\lambda), \quad \psi_{-v}(\lambda) = -\psi_v(-\lambda). \quad (3)$$

Convexity of ψ_v makes $\lambda \mapsto \psi_v(-\lambda)$ convex. Convexity of the negative of this function, supplied by ψ_{-v} , makes it concave. Hence ψ_v is simultaneously convex and concave and is therefore affine, say $\psi_v(\lambda) = a_v + b_v \lambda$. Consequently

$$K_v(\lambda) = a_v \lambda + b_v \lambda^2.$$

Differentiating at zero gives $a_v = \mathbb{E} \langle v, X \rangle$ and $2b_v = \text{Var}(\langle v, X \rangle) \geq 0$. This is exactly the cumulant generating function of $N(a_v, 2b_v)$; uniqueness of moment generating functions proves the assertion. $\square$

Theorem 1 (Universal Gaussian convexity characterizes Gaussian laws). *Under (1), the following are equivalent.*

1. *For every finite convex $F : \mathbb{R}^n \rightarrow \mathbb{R}$ with $\mathbb{E} e^{\lambda F(X)} < \infty$ for every $\lambda > 0$, the function $\Lambda_{F,X}$ is convex on $\mathbb{R}$.*
2. *The same assertion holds when F is restricted to affine functions.*
3. *X is multivariate Gaussian, with arbitrary mean and positive semidefinite covariance; singular and deterministic laws are allowed.*

Proof. The implication (1) $\Rightarrow$ (2) is immediate. Assume (2). For every $v \in \mathbb{R}^n$, apply it to the two linear convex functions $x \mapsto \langle v, x \rangle$ and $x \mapsto -\langle v, x \rangle$. Lemma 1 shows that every one-dimensional projection $\langle v, X \rangle$ is Gaussian. By the standard projection characterization (equivalently, by evaluating the characteristic function on every v), X is a multivariate Gaussian. This proves (2) $\Rightarrow$ (3).

Conversely, write a general Gaussian vector as $X = m + Ag$, where g is a standard Gaussian vector of some finite dimension and A may be singular. For convex F , the composition $z \mapsto F(m + Az)$ is convex. Chen's theorem applied to this composition gives convexity of $\Lambda_{F,X}$, proving (3) $\Rightarrow$ (1). The deterministic case corresponds to $A = 0$ and is included. $\square$

Corollary 1 (Minimal test class). *Among laws satisfying (1), it is enough to verify the convexity for the $2n$ coordinate tests $F(x) = x_j$ and $F(x) = -x_j$ together with their analogues after every linear change of coordinates. Equivalently, the paired tests $F(x) = \pm \langle v, x \rangle$ for all v are complete.*

4 An explicit obstruction and sharpness of the scope

Let X be a symmetric Rademacher random variable. The identity test is convex, but its normalized log-mgf is

$$\Lambda_X(\lambda) = \frac{\log \cosh \lambda}{\lambda}, \quad \Lambda_X(0) = 0.$$

It is not convex on $\mathbb{R}$. Indeed,

$$\Lambda_X(\lambda) = \frac{\lambda}{2} - \frac{\lambda^3}{12} + \frac{\lambda^5}{45} + O(\lambda^7),$$

and directly $\Lambda_X''(1) \approx -0.235652 < 0$ (while its value at -1 has the opposite sign). Thus even the simplest bounded non-Gaussian law does not support a blanket extension.

The quantifier over *all* convex observations is essential to the classification. Fernández-Unzueta, Melbourne, and Palafox-Castillo [2] subsequently studied scalar variables whose own normalized log-mgf is convex and proved strict convexity for a broad Ehrhard-like class. That phenomenon does not contradict Theorem 1: a non-Gaussian variable may pass one observation, while the paired affine family forces a universal carrier law to be Gaussian.

5 Verification, novelty, and verdict

The included symbolic script checks the Gaussian affine formula, the Rademacher series, and the signs of its second derivative at ± 1 . These checks are ancillary; the classification proof is the two-line reflection identity (3) followed by the projection characterization.

A bounded novelty search on 17 August 2026 covered the run’s four cheap indexes and web searches for the source title, its extension phrase, and combinations of universal convex tests with Gaussian characterization. The most relevant later paper was [2]; it develops the scalar fixed-variable direction and a strictness theorem, but does not state the universal carrier classification above. No prior statement of this exact iff result was found. Novelty is plausible, not certified.

This packet gives a full answer to the natural universal interpretation of the source’s question. It does not classify laws for which convexity holds for one prescribed nonlinear F , nor intermediate restricted families of tests. Human review should audit the intended quantifier in the source question and the mild all-linear-exponential-moments hypothesis. Subject to that interpretation, the proof is complete.

References

- [1] W.-K. Chen, *A Gaussian Convexity for Logarithmic Moment Generating Functions with Applications in Spin Glasses*, arXiv:2311.08351v2, 2024.
- [2] M. Fernández-Unzueta, J. Melbourne, and G. Palafox-Castillo, *On Convex Functions of Gaussian Variables*, arXiv:2510.06676, 2025.